

Preliminary malacological investigation on the loess profile at Zmajevac, Croatia

Communication

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Abstract: More than 5 500 specimens of 37 terrestrial snail species were collected from a loess paleosol section at Zmajevac village, Croatia. The presence of *Ena montana*, *Mastus bielzi*, *Cochlodina laminata*, *Macrogastra ventricosa*, *Clausilia pumila*, *Clausilia dubia*, *Pseudofusus varians*, *Trichia unidentata* and *Trichia edentula* species from the lower paleosol layer suggests that this malacological zone formed during the initial phase of a Middle Pleistocene interglacial cycle.

Keywords: loess profile • quartermalacology • mollusc • Croatia • Zmajevac village

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1. Introduction

The quartermalacological sampling of the loess profile at Zmajevac was carried out by a Croatian-Hungarian collective collaboration in 2008. This collaboration was organized to study loess profiles in SW-Hungary and north Croatia, located at Šarengrad and Zmajevac respectively.

The leader of the researcher-team was Dr. Lidija Galović (Croatian Geological Survey) who already examined this profile in her PhD thesis. The other members of the research team were Dr. Géza Chikán and Dr. László Koloszár (Hungarian Geological Institute), Dr. Pál Sümegi and Sándor Gulyás (Department of Geology and Paleontology, University of Szeged, Hungary).

The examined loess profile is situated east of the village of Zmajevac (Veresmart), in the southern part of Baranja County, Croatia (N: 45°48'44.84"; E: 18°49'13.60"). The profile is 26 meters high. Due to financial and temporal constraints only 3 pre-selected parts of the entire section were sampled in this study. These 3 parts were between 24-25 m, 11-13 m and 1-5.5 m.

2. Sampling methods

From the cleaned profile sections the research team undertook the following methods to retrieve samples conforming to the quartermalacological sampling method: One cubic-decimeter (about 2.5 kg) was taken from every 25 cm of the profile [1, 2]. The samples were wet screened using a screen of 0.5 mm mesh. A total of 31 samples were collected and the shells were taxonomically identified.

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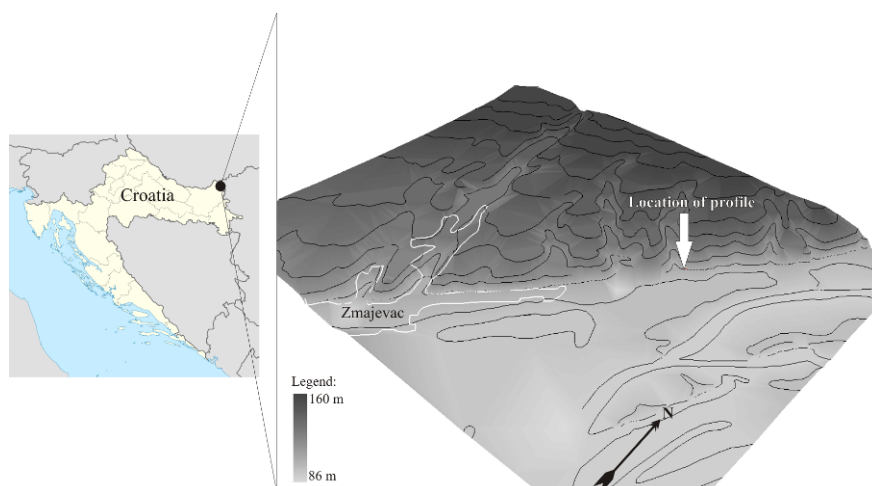


Figure 1. The location of the loess profile at Zmajevac.

The abundance and dominance values of taxonomically identified specimens were noted for each profile, and charts prepared to depict the specimen and percentage distribution of the individual species with the depth of profile. Next, the individual species were clustered into paleoecological and biogeographical groups. To do this, the species ecological requirements were taken into account, such as preferred temperature, humidity or vegetation cover [3–5]. Fluctuations in the abundance and dominance values of the individual paleoecological and biogeographical groups with depth are depicted in every profile, with charts and tables prepared with the Psimpoll software package of Bennett (1992). For a color description of the profile the Munsell Soil Color Charts were used.

Climatic reconstructions were carried out using the malacothermometer method. This method is based on the areas current biogeographic patterns of 11 dominant gastropod species from a composite malacofauna. For selected gastropod species, the optimal climatic conditions can be determined, along with the minimum and maximum temperature values of tolerance (the activity range of gastropods), with the help of data from meteorological stations [4, 6, 7].

3. Environmental and stratigraphic description

The profile at Zmajevac can be found along a road near the floodplain of the river Danube, on the loess covered southern part of the 245–251 m high, island-like Banske Hills, which is mainly composed of Miocene basalt and andesite (see Figure 1).

A mass of empty but modern shells of *Pomatias elegans*, *Helix pomatia* and *Helicella obiva* were found on the surface in this area. On the surroundings of the examined loess profile there were orchards, farmed and dropped vineyards, almond and walnut plantations. The margin of the loess-wall is covered by silver lime (*Tilia tomentosa*), black locust (*Robinia pseudoacacia*), field elm (*Ulmus minor*), with natural-like steppe stretches, black elder (*Sambucus nigra*), *Euonymus verrucosus* and wayfaring tree (*Viburnum lantana*). There were also hellebor (*Helleborus odorus*), greater celandine (*Chelidonium majus*), ivy-leaved speedwell (*Veronica hederifolia*), goosegrass (*Galium aparine*) and wood avens (*Geum urbanum*) from the non-arboreal plants.

The bedrock of the profile consists of aeolian loess (light yellowish brown colour 2.5 Y 6/4). Above the loess bedrock, between 23 and 26 m, there is a polygenetic paleosol complex, with about a meter thick bench-like carbonate facies and concretions in its lower part (see Figure 2). At least 3 different soil horizons could be distinguished in the paleosol complex based on sedimentary texture. Above the carbonated layer there is a meter thick soil level BC (5 Y 7/4 pale yellow – 10 YR 5/4–6/4 yellowish and light yellowish brown) which is a transitional horizon from the bedrock. In this layer (between 24 and 25 m) there is a mildly laminated, redeposited sediment level, infilling with a thin erosional discordance. This zone is highly fossiliferous. Above there is a dark brown, meter thick, organic rich paleosol horizon B with iron-nodules, and a yellowish brown, half meter-thick humic level which is a paleosol horizon A. On the top of the soil complex a 4 m thick (2.5 Y 7/2 – light gray) mildly laminated slope loess layer was deposited. No samples were taken from

this loess horizon.

On the top of the slope loess layer there is a 5 m thickness of redeposited paleosols and sediments, which is termed a pedosediment layer. The sediment has different textures, from a mildly laminated yellowish brown texture to a dark grey waved texture [8]. On the top of this pedosediment layer a 2 m thick, well sorted, calcareous and fossiliferous (2.5 Y 6/4 – light yellowish brown) aeolian loess is deposited. Samples were taken from from this loess layer along a gorge. On the top of the loess layer a double soil complex formed, which is about 2 m thick and contains the fossil soil's B and BC horizons (10 YR 4/4-6/4 – dark and light yellowish brown). On this complex is a 2 m thick, faintly laminated, calcareous, non-fossiliferous aeolian loess layer. On the top of this loess layer is an other double paleosol complex (with B and BC horizons), which is about 2 m thick (10 YR 4/4-6/4). The repetition of this layer shows that this complex is a recurrence of the loess-soil layer in the profile's bedrock. This layer-structured series of sediments generally develops on pedimental areas, but probably on foothills too, because of the along slope gapping of the loess and soil layers. The upper 6 m of the profile contains loess layers with a significant mollusc shell-content, which is suitable for stratigraphic and paleoecological examinations.

4. The mollusc fauna

The studied profiles yielded 5 501 specimens of 37 mollusc species. The most significant mollusc fauna is from the lower part of the profile, from 24 to 25 m. This paleosol complex yielded a number of *Cochlodina laminata*, *Clausilia pumila*, *Macrogastra ventricosa*, *Pseudofusulus varians*, *Mastus bielzi*, *Orcula dolium*, *Aegopinella ressmanni*, *Semilimax semilimax*, *Ena montana*, *Trichia unidentata* and *Trichia edentula* species. These species are commonly found in temperate deciduous forests today (see Figure 3).

As the terrestrial loess areas of Croatia have not processed yet, the quaternary malacofauna, which are present in the lower parts of the Zmajevac profile, are be analyzed without comparisons to living species. However, the common occurrence of *Ena montana*, *Mastus bielzi*, *Trichia unidentata* and *Trichia edentula* species clearly demonstrates that the malacofauna and the fauna-bearing soil layer were developed in the Middle Pleistocene [1, 9]. As the polygenetic soil-sediment complex is covered by a meter thick sediment complex, it can be proposed that the redeposited sediment was probably formed in the Middle Pleistocene.

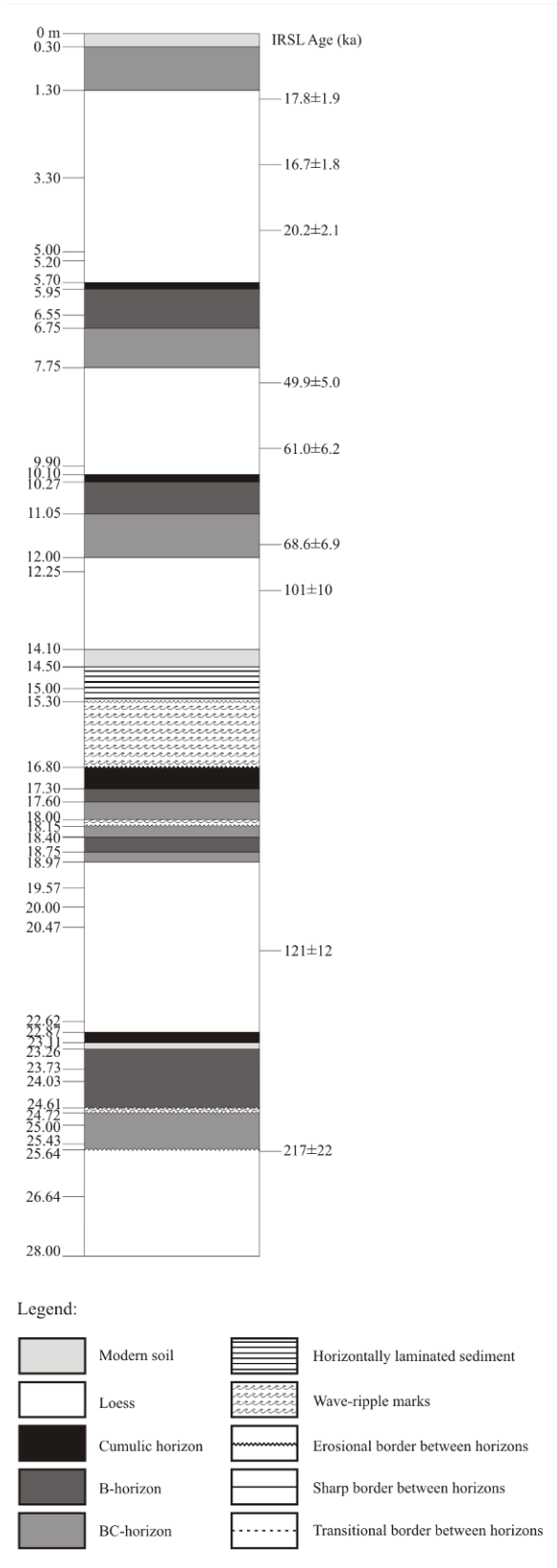


Figure 2. A detailed sketch of the loess profile with IRSL Ages [8].

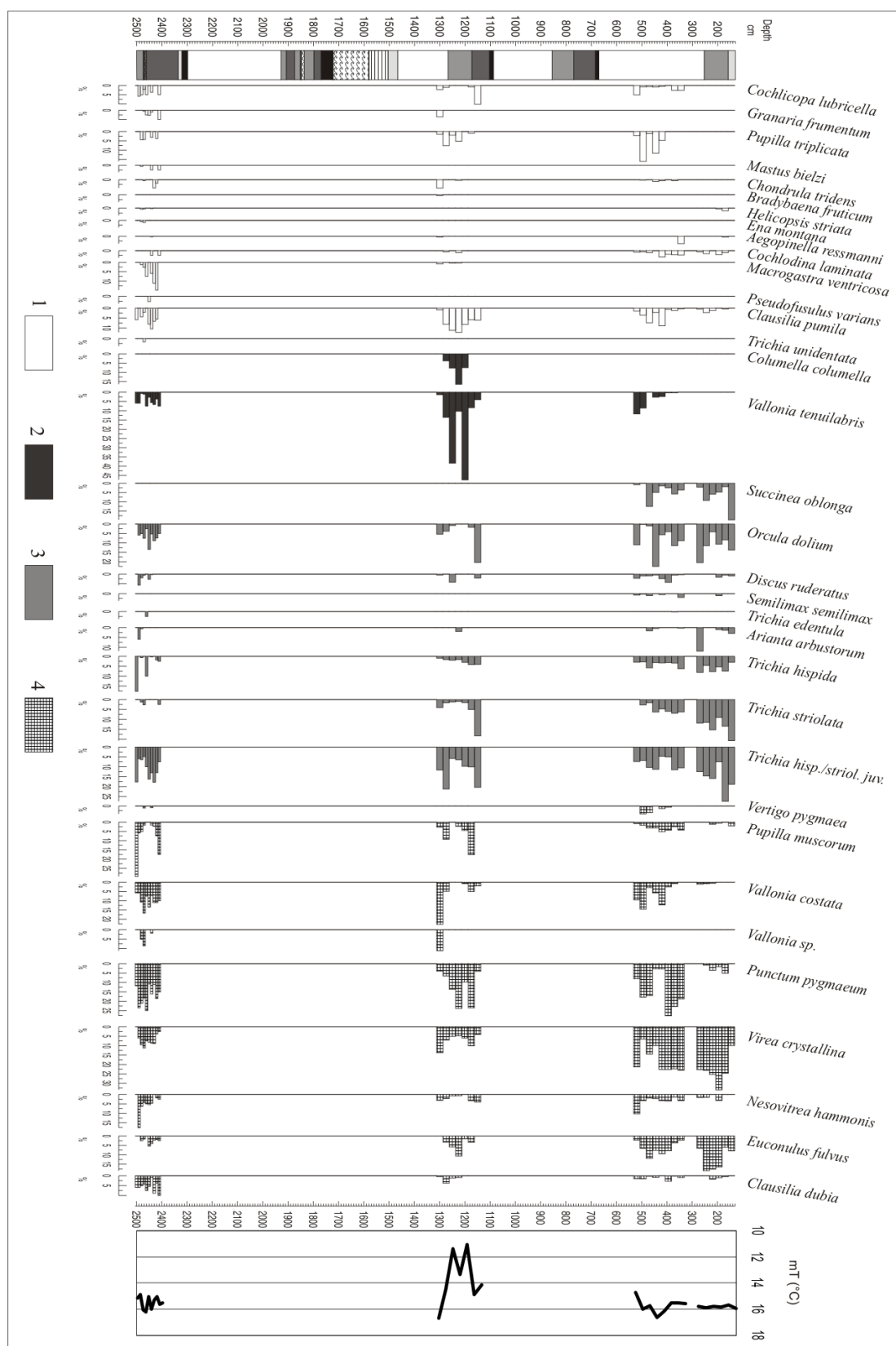


Figure 3. The dominance relationships of the mollusc species from the profile along with malacothermometer data (1. thermophilous, 2. cryophilous, 3. cold resistant, 4. mesophilous).

The abundance of the *Discus ruderatus* species indicates that deciduous forest may have been mixed with conifers. Additionally, the presence of *Trichia*, *Arianta arbustorum* and *Bradybaena fruticum* species means that at the site there were gallery forests with moist herb layers and meadows. The implication of the presence of *Vallonia*, *Punctum pygmaeum*, *Nesovitreia hammonis*, *Vitreia crystallina* and *Euconulus fulvus* species is that the edge of wood had a well developed shrub stratum within a closed and species-rich ecotone zone.

This environmental interpretation has only one contradiction: the presence of the *Vallonia tenuilabris*, as the temperature range of xeromesophilous steppe-species (*Pupilla muscorum*, *Chondrula tridens*, *Pupilla triplicata*, *Vertigo pygmaea*) does not deviate from forest species too much. Based on Quaternary paleobotanical and malacological research [9–11], this sediment layer developed in a mixed leaved forest-steppe environment which formed at the beginning or terminal phase of a Middle Pleistocene interstadial climatic cycle. In this environment all of the mentioned species could occupy the ideal niche for themselves, so the collective presence of these species is not a surprise. However, the Recent and Upper Pleistocene temperature range of *Vallonia tenuilabris* notably deviates from the other species (listed above). This anomaly suggests that the evolution of this part of the profile may have undergone special taphonomic processes, which needs to be examined further in the future.

Of quatermalacological importance is the presence of the *Pseudofusus varians* species in this part of the loess profile. Although fossil localities of this species are known in Central Europe, mainly in foothills region of the Alps and in the Czech Basin, in the Carpathian Basin this species is only extremely rarely found in Pleistocene sediments.

5. Discussion

The consistence and dominance relationships of the fauna and the stratigraphic setting suggest that this part of the loess profile at Zmajevac developed in a terminal period of an interglacial period, in the Middle Pleistocene. Probably it is a younger horizon than the time of the Mindel glacial, because there are no exotic, subtropical species, which were common in the early period of the Middle Pleistocene [1, 2]. It is possible that this horizon could be from the early part of the Mindel/Riss interglacial. However, this cannot be confirmed by the available evidence from malacological and chronological data. From the available data it can only securely be stated that this horizon developed in the later period of the Middle Pleistocene, at an early period of an interglacial, between ap-

proximately 350 000 and 130 000 years [8].

With further sampling and paleoecological examinations, more detailed paleoecological and malacological data may be available.

References

- [1] Krolopp E., Quaternary malacology in Hungary, Földrajzi Közlemények, 1973, 21, 161–171
- [2] Krolopp E., Biostratigraphic division of Hungarian Pleistocene Formations according to their Mollusc fauna, Acta Geologica Hungarica, 1983, 26, 69–82
- [3] Krolopp E., Sümei P., Paleocological reconstruction of the Late Pleistocene, based on Loess Malacofauna in Hungary, GeoJurnal, 1995, 36, 213–222
- [4] Sümei P., Loess and Upper Paleolithic environment in Hungary, Aurea Press, Nagykovácsi, 2005
- [5] Sümei P., Krolopp E., Quatermalacological analyses for modelling of the Upper Weichselian paleoenvironmental changes in the Carpathian basin, Quatern. Int., 2002, 91, 53–63
- [6] Sümei P., Hajdúság felső-pleisztocén fejlődéstörténete finomrétegtani (üledéköltani, őslénytani, geokémiai) vizsgálatok alapján, PhD thesis, University of Debrecen, Hungary, 1989 (in Hungarian)
- [7] Sümei P., Az ÉK-magyarországi löszterületek összehasonlító őskörnyezeti rekonstrukciója és rétegtani értékelése. Candidate thesis, University of Debrecen, Hungary, 1996 (in Hungarian)
- [8] Galović L., Frechen M., Halamić J., Durn G., Romić M., Loess chronostratigraphy in Eastern Croatia – A luminescence dating approach, Quatern. Int., 2009, 198, 85–97
- [9] Ložek V., Quartärmollusken der Tschechoslowakei, Rozpravi Ústředního ústavu geologického, 1964, 31, 374
- [10] Marković S. B., Bokhorst M. P., Vandenberghe J., McCoy W. D., Oches E., Hambach U., Gaudenyi T., Jovanović M., Zöller L., Stevens T., Machalet B., Late Pleistocene loess-paleosol sequences in the Vojvodina region, north Serbia, J. Quaternary Sci., 2008, 23, 73–84
- [11] Marković S. B., Oches E., Sümei P., Jovanović M., Gaudenyi T., An introduction to the Middle and Upper Pleistocene loess-paleosol sequence at Ruma brickyard, Vojvodina, Serbia, Quatern. Int., 2006, 149, 80–86